The Lucas Critique Examined in ML

D200 Machine Learning in Economics

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Outline

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The Lucas Critique (1976)

Core Argument

Parameters of reduced-form econometric models are not policy-invariant. When a policy changes, agents update their expectations, altering the model's coefficients and rendering pre-change forecasts invalid.

Formally, if an agent's rule is $x_t = f(y_t; \theta)$ where θ embeds beliefs about policy $\{\pi_t\}$, then a shift $\pi \to \pi'$ forces $\theta \to \theta'$ invalidating any forecast built on $\hat{\theta}$.

Classical response: VARs, regime-switching models reduce theory dependence while retaining parametric structure.

ML: Model-agnostic, data-driven, flexible functional form. Do ML models provide a hedge against the Lucas Critique, or does over-fitting to the pre-break regime make things *worse*?

Research Question

Are regime-switching ML models more or less susceptible to the Lucas Critique than classical parametric alternatives, and **why**?

Paper's Headline

ML models degrade *similarly* to classical models but still produce **better out-of-sample forecasts** under regime change.

Why This Matters

- Rapidly shifting macroeconomic regimes inflation shocks, financial crises, unconventional monetary policy are the norm. Forecasting under regime change is directly relevant to policy-makers, central banks, and institutions using ML.
- The Lucas Critique motivated the rise of VARs in the 1980s and DSGE micro-foundations yet model instability under structural breaks remains an open problem for *all* model classes, parametric or ML.
- ML's flexible functional form (Athey, 2018) may reduce *approximation error* post-break. But greater in-sample flexibility may *increase estimation error* when the distribution shifts a fundamental tension.
- Connects to Breiman's "two cultures" and Box's "all models are wrong": whether statistical modelling or algorithmic prediction better handles regime instability is active and unresolved.

What Is Already Known

Classical response to Lucas:

- **VARs** (Bormetti & Corsi, 2022) reduced-form, model-agnostic, no structural identification.
- **TVP-VARs** (Primiceri, 2005) time-varying parameters; still parametric.
- **MSM / Hamilton filter** (1989) latent regimes; parameters vary by regime, not by policy change.
- **DSGEs** micro-founded; Storm (2021) argues still vulnerable to structural misspecification.

ML and the Lucas Critique:

- **Athey (2018)** data-driven model selection; may outperform parametric models when structure shifts.
- **Chen (2018)** ML on Progresa programme; relative forecasting strength under changing structure.
- **Salle (2015)** ANN expectation formation; Lucas-robust as no structural form assumed.
- **Pereira (2025)** ML rethinks econometric methodology (predictive economics manifesto).

Gap This Project Fills

No systematic *head-to-head* comparison of classical and ML regime-switching models on the same Lucas Critique experiment with both simulated (known DGP) and real macroeconomic data.

Data: Two Experimental Settings

Experiment A Simulated DGP

Two-state Markov-switching AR(1) with known ground truth:

$$y_t = \mu_{s_t} + \phi_{s_t} y_{t-1} + \sigma_{s_t} \varepsilon_t, \quad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$$

	Mild		Severe	
Param	Pre	Post	Pre	Post
μ_0	0.5	0.2	0.5	-0.5
μ_1	-1.5	-1.0	-1.5	-3.0
σ_0	1.0	1.3	1.0	2.0
σ_1	2.5	3.0	2.5	4.5
P_{00}	0.95	0.90	0.95	0.80

Mild = gradual tightening; Severe = Volcker-scale

Experiment B Real-world FRED data

Series	Break	Period
INDPRO	Great Moderation	Jan 1984
CPIAUCSL	Volcker disinflation	Oct 1979

Covariates: EFFR, UNRATE, USREC (NBER).

Transformations: log-differences $\times 100$; lags; 5/20-period rolling mean/std; trend-cycle decomposition for HMM features.

Protocol: Train pre-break only; evaluate post-break. No re-training or look-ahead. Models frozen at pre-break estimates.

Eight Models: A Taxonomy

Model	Family	Regime identification	Forecast
AR(2)	Linear baseline	None	AR linear
ARMA(2,1)	Linear baseline	None	AR+MA linear
MSM	Classical reg.-switch	Hamilton filter (EM)	Regime-wtd AR
TAR	Classical reg.-switch	Threshold on y_{t-1}	Per-regime OLS
HMM + Ridge	Probabilistic ML	Baum-Welch EM	Per-regime Ridge
Mixture of Experts	Probabilistic ML	Soft gating via EM	Wtd Ridge
ML Regime (XGB)	Nonlinear ML	KMeans $\rightarrow$ XGBClassifier	Per-regime XGBRegressor
MSNN	Nonlinear ML	Markov chain + MLP (EM)	Per-regime MLP
Model Average	Ensemble		Equal-weight mean

Evaluation Metric Lucas Sensitivity Ratio (LSR)

$$\mathrm{LSR} = \frac{\mathrm{RMSE}_{\mathrm{post}}}{\mathrm{RMSE}_{\mathrm{pre}}} \qquad \begin{cases} \mathrm{LSR} > 1 & \text{Lucas-critique vulnerable (post-break worse)} \\ \mathrm{LSR} < 1 & \text{inverse Lucas: variance-reducing break (post-break better)} \end{cases}$$

Theoretical Framework: Forecast Risk Decomposition

DGP: $y_{t+1} = f_{s_t}(x_t) + \varepsilon_{t+1}$, $s_t \in \{1, \ldots, S\}$. Optimal forecast: $m^*(x_t) = \sum_s \Pr(s_t{=}s \mid x_t)\, \mathbb{E}[y_{t+1} \mid x_t, s_t{=}s]$.

Total forecast risk decomposes as:

$$\mathbb{E}\big[(y_{t+1} - \hat{m}(x_t))^2\big] = \underbrace{\sigma^2}_{\text{irreducible}} + \underbrace{\mathbb{E}\big[(m^* - m(\cdot;\mathcal{F}))^2\big]}_{\text{approximation error}} + \underbrace{\mathbb{E}\big[(m(\cdot;\mathcal{F}) - \hat{m})^2\big]}_{\text{estimation error}}.$$

Under the Lucas Critique:

- Parametric MSM: low estimation error (correct spec.) but high *approximation* error once DGP shifts.
- Flexible ML: low approximation error in-sample but high *estimation* error post-break (over-fitting).
- Variance-reducing break (INDPRO 1984): σ^2 falls $\Rightarrow$ LSR < 1 even with frozen $\hat{\theta}$ *inverse Lucas Critique*.

Key Prediction

MSM minimises estimation error under the correct spec. but collapses when the DGP shifts. ML does the opposite. Neither class is immune.

Ensemble insight

Expanding $\mathcal{F}$ via model averaging reduces approximation error theoretical basis for the ensemble's stability.

HMM + Ridge: Two-Stage Pipeline

Stage 1 Gaussian HMM (Baum-Welch EM):
$\mathbf{O}_t = [y_t,\ y_{t-1},\ \bar{y}_t^{(5)},\ s_t^{(5)},\ \bar{y}_t^{(20)}]^\top$
Short/long-run trend-volatility summaries help detect transitions. Viterbi decoding gives $s^*_{1:T}$; causal forward filter at test time.
Stage 2 Per-regime Ridge: $\hat{y}_t = \hat{\beta}_{s_t^*}^\top x_t$,
$x_t = [y_{t-1},\ \bar{y}_t^{(5)},\ s_t^{(5)},\ \text{EFFR},\ \text{UNRATE}]$

```
O = df[["y","y_lag1","roll_mean_5",
        "roll_std_5","roll_mean_20"]].values
hmm = GaussianHMM(n_components=2, n_iter=200).fit(O)
regimes = hmm.predict(O)        # Viterbi
for k in range(2):
    mask = (regimes == k)
    regressors[k] = Ridge(alpha=1.0).fit(X[mask], y[mask
        ])
# Test: causal forward filter (no look-ahead)
regime = forward_filter_predict(O_new)
yhat   = regressors[regime].predict(x_new)
```

Why HMM is most robust

Soft regime assignments adapt partially to a shifted emission distribution unlike hard KMeans frozen boundaries.

Ridge regressors resist overfitting.

LSR = **1.51** (mild), **2.29** (severe) best across all 4 experiments (simulated + real-world).

The Viterbi Algorithm: Decoding Hidden States

Goal: find the most probable hidden-state sequence:
$s^*_{1:T} = \arg\max_{s_{1:T}} P(s_{1:T} \mid \mathbf{O}_{1:T})$

Dynamic programming in log-space (K states, T steps):

1. **Initialise:** $\log \delta_1(k) = \log \pi_k + \log b_k(\mathbf{O}_1)$
2. **Recurse** $t = 2, \ldots, T$, storing backpointers $\psi_t(k)$:
$\log \delta_t(k) = \max_j [\log \delta_{t-1}(j) + \log A_{jk}] + \log b_k(\mathbf{O}_t)$
3. **Backtrack** from $s^*_T = \arg\max_k \log \delta_T(k)$:
$s^*_t = \psi_{t+1}(s^*_{t+1}), \quad t = T-1, \ldots, 1$

	Viterbi	**Fwd-bwd smoother**
Output	hard $s^*_t \in \{0, 1\}$	soft $\gamma_t(k) \in (0, 1)$
Operation	max over paths	sum over paths
Training use	decoding (HMM)	E-step (MSNN)

HMM + Ridge

- *Train:* Viterbi decodes $s^*_{1:T} \to$ each regime subset trains a Ridge regressor.
- *Forecast:* causal **forward filter** (not Viterbi) conditions only on past observations; no look-ahead.

MSNN

- *Train (E-step):* **forward-backward smoother** computes soft $\gamma_t(k)$ for weighted MLP updates. Viterbi is *not* used during training.
- *Forecast:* extend Viterbi path one step to select the current regime's MLP expert.

Hard vs. soft: the Lucas-stability link

Viterbi gives a hard assignment (one state per t). The **forward-backward smoother** gives soft $\gamma_t(k) \in (0, 1)$ letting HMM partially adapt to a shifted emission distribution *without re-training*. This is why HMM achieves the lowest LSR in all four experiments.

ML Regime (XGB) & Mixture of Experts

ML Regime three stages:

1. KMeans on lags + rolling stats → pseudo-regime labels
2. XGBClassifier reproduces labels from same features
3. Per-regime XGBRegressor trained on labelled subset

```
km     = KMeans(n_clusters=2).fit(X)
labels = km.labels_
clf    = XGBClassifier(n_estimators=200)
clf.fit(X, labels)
for k in range(2):
    regs[k] = XGBRegressor().fit(
        X[labels==k], y[labels==k])
```

Vulnerability: KMeans boundaries are frozen. XGB's piecewise-constant leaves memorise training data perfectly but extrapolate poorly leaf values become systematically wrong post-break.
Pre-RMSE = **0.396** (best); LSR = **8.43** (worst).

Mixture of Experts soft gating via EM:

$K = 2$ Ridge regressors with a logistic gating network. Regime probability updated by EM:

$$R_{tk} = P(\text{expert} = k \mid x_t, y_t) \propto \pi_k(x_t) \cdot \mathcal{N}(y_t;\, \hat{\mu}_{kt}, \hat{\sigma}_k^2)$$

The EM alternates between updating R_{tk} and refitting the gating network + Ridge experts. Forecast:

$$\hat{y}_t = \sum_k \pi_k(x_t) \cdot \hat{\beta}_k^\top x_t$$

Unlike TAR or XGB, each expert contributes proportionally to its gating weight partial regularisation, reduced sensitivity to regime misclassification. LSR = 1.96 (mild shift).

MSNN: Markov Switching Neural Network

Extends Hamilton's MSM by replacing linear regime-specific models with MLP experts, estimated jointly via EM.

E-step log-space Hamilton forward-backward filter:

$$\gamma_t(k) = P(s_t{=}k \mid y_{1:T}), \quad \xi_t(j,k) = P(s_t{=}j, s_{t+1}{=}k \mid y_{1:T})$$

M-step analytical transition update + Adam for MLPs:

$$\hat{P}_{jk} \propto \textstyle\sum_t \xi_t(j,k), \quad \mathcal{L}_k = \textstyle\sum_t \gamma_t(k)(y_t - f_k(x_t))^2$$

```
# E-step: log-space forward filter
log_alpha = log_pi + log_emit(y[0])
for t in range(1, T):
    log_alpha = logsumexp(
        log_alpha[:,None]+log_A, axis=0
    ) + log_emit(y[t])
# M-step: weighted MLP update
for k in range(K):
    mlp[k].fit(X, y, sample_weight=gamma[:,k])
```

MLP architecture: $n_{\text{feat}} \to 32 \to 16 \to 1$, ReLU, linear output, Adam ($\eta = 3{\times}10^{-3}$), 200 epochs.

Key advantage: Smooth MLP interpolation degrades gracefully post-break unlike XGB's step functions.

Key weakness: EM collapses to one dominant regime in-sample (accuracy $\approx 51\%$; Markov constraint not binding).

Despite this: **best absolute post-break RMSE** (2.493 mild, 3.391 severe) smooth extrapolation dominates the regime failure.

Simulated Results: Lucas Sensitivity Ratios

	Simulated Mild			Simulated Severe		
Model	Pre	Post	LSR	Pre	Post	LSR
AR(2) Baseline	1.634	2.706	1.66	1.634	4.261	2.61
ARMA(2,1)	1.674	5.577	3.33	1.674	11.61	3.33
MSM	1.673	5.968	3.57	1.673	11.96	7.15
TAR	1.537	2.526	1.64	1.537	4.266	2.78
HMM	1.696	2.559	**1.51**	1.696	3.892	**2.29**
Mixture of Experts	1.339	2.628	1.96	1.339	4.742	3.54
ML Regime (XGB)	0.396	3.337	8.43	0.396	8.471	21.41
MSNN	0.613	**2.493**	4.07	0.613	**3.391**	5.53
Model Average	1.222	<u>2.015</u>	2.47	1.222	6.007	4.92

By LSR (relative degradation)

HMM best (1.51), **XGB** worst (8.43). MSM *worse* than AR(2) Hamilton filter collapses post-break.

By absolute post-break RMSE

MSNN best (2.493) despite high LSR. Rankings invert both metrics are needed.

Real-World Results: INDPRO & CPI

Industrial Production Great Moderation (1984)

Model	Pre	Post	LSR
AR(2)	0.717	0.514	0.72
MSM	0.823	0.723	0.88
TAR	0.624	0.397	0.64
HMM	0.648	**0.378**	**0.58**
MoE	0.619	0.407	0.66
XGB	0.140	0.443	3.18
MSNN	0.326	0.493	1.51

Inverse Lucas: INDPRO std dev halves ($0.892 \rightarrow 0.512$), falling σ^2 improves accuracy for 7/9 models. Only XGB and MSNN degrade (pre-break overfitting).

CPI Inflation Volcker Disinflation (1979)

Model	Pre	Post	LSR
AR(2)	2.631	2.594	0.99
MSM	2.954	4.038	1.37
TAR	1.912	2.276	1.19
HMM	2.040	2.311	**1.13**
MoE	1.791	2.992	1.67
XGB	0.357	2.559	7.18
MSNN	1.852	3.068	1.66

Classic Lucas pattern: all models degrade. HMM most stable (LSR=1.13). Chow *fails* ($F = 1.39$, $p = 0.25$) Volcker is a shift in regime *transition dynamics*, not AR parameters.

Cross-Experiment Summary

Finding	Sim (Mild)	Sim (Severe)	INDPRO 1984	CPI 1979
All models LSR > 1?	Yes	Yes	No 7/9 improve	Yes
XGB worst LSR?	Yes (8.4)	Yes (21.4)	Yes (3.2)	Yes (7.2)
HMM most stable?	Yes (1.51)	Yes (2.29)	Yes (0.58)	Yes (1.13)
Chow detects break?	Yes (F=13.7)	Yes	Yes (F=5.83)	No (F=1.39)

Cross-experiment invariant

XGB is most vulnerable in every experiment. **HMM** is most robust. These rankings are preserved regardless of break type, direction, simulated or real.

The dual-metric tension

HMM = best LSR. **MSNN** = best absolute post-break RMSE. **XGB** = best pre-break RMSE *and* worst LSR. The metric chosen determines the model ranking.

What Worked & Why

HMM best LSR all 4 experiments

- *Flexible latent-state estimation* (soft posteriors), not functional form probabilistic assignments adapt to shifted emission distributions without re-training.
- Ridge regressors resist overfitting to pre-break noise. LSR = 1.51 (mild), 2.29 (severe).

MSNN best absolute post-break RMSE

- Smooth MLP extrapolation degrades more gracefully than XGB's step functions. Advantage grows with shift magnitude (2.493 mild, 3.391 severe).

Model average stable ensemble

- Lower post-break RMSE *and* lower LSR in 2/4 experiments. Expanding $\mathcal{F}$ reduces approximation error (consistent with the risk decomposition).

XGB catastrophic failure

- Piecewise-constant leaves memorise pre-break regime structure; post-break misclassification causes systematic leaf-value errors.
- Pre-break RMSE 0.40: a *denominator effect* inflating LSR to 8.43 (mild) / 21.41 (severe). Frozen KMeans regime classifier is most brittle under distributional shift.

MSM correctly specified, then collapses

- Correctly-specified for the DGP. Fixed EM estimates $(\hat{\mu}_k, \hat{\sigma}_k, \hat{P})$ cause Hamilton filter collapse: regime accuracy 75%→20%, ARI→0.
- LSR = 3.57 *worse* than AR(2) (1.66). Correct pre-break specification *increases* Lucas vulnerability.

Key Insight

Models flexible in *latent-state estimation* (HMM) outperform those flexible in *functional form* (XGB) under structural breaks.

What Did Not Work, Limitations & Hypotheses

Limitations:

- **No online updating** by design (isolates the Lucas mechanism), but limits practical relevance. Key next step.
- **MSNN regime alignment poor** accuracy $\approx 51\%$ (near random); EM collapsed to one dominant regime; Markov constraint not binding.
- **CUSUM detects break 156 periods late** under mild shift; limited real-time usefulness.
- **Chow test misses Volcker break** dynamics shift, not parameter shift; Chow cannot detect it.
- **LSR metric-sensitive** to pre-break RMSE; cannot compare across models with very different in-sample fit.
- Single train/test split for real-world data limits power.

Formal hypothesis verdicts:

H	Result
H1: all LSR>1	Not rejected for variance-increasing breaks
H2: regime $<$ AR	Rejected HMM wins but MSM $>$ AR(2)
H3: ML over-fits	Confirmed for XGB; partial for MSNN
H4: break direction	Confirmed INDPRO inverse Lucas

Overall verdict

ML does *not* resolve the Lucas Critique. Flexible *latent-state* models (HMM) beat flexible *functional form* models (XGB) under structural breaks.

Next Steps

1. **Online / sequential updating.** Rolling-window or Bayesian online learning: how quickly does each model class recover post-break? This directly tests the practical Lucas Critique response.
2. **Composite stability metric.** Combine LSR and absolute post-break RMSE to avoid the denominator-effect pathology: $\mathrm{Composite} = \alpha \cdot \mathrm{LSR} + (1-\alpha) \cdot \mathrm{RMSE}_{\mathsf{post}}/\bar{\sigma}$.
3. **Richer DGP and more real-world series.** Non-linear DGPs (GARCH-regime, threshold GARCH), additional FRED series, multi-country panels testing generalisability of the HMM/XGB ranking.
4. **Optimised ensembles.** The model average already outperforms in 2/4 experiments. Stacking or Bayesian model averaging with Lucas-stability priors could improve further.
5. **Attention-based sequence models.** Transformers implicitly re-weight past observations this may confer natural Lucas robustness without explicit regime modelling.

Summary

Main Findings

1. **Lucas Critique affects all classes** no model is immune under variance-increasing breaks.
2. **HMM most stable** (best LSR all 4 experiments) flexible *latent-state estimation*, not functional form, is key.
3. **XGB catastrophically vulnerable** best pre-break RMSE, worst LSR in every experiment.
4. **Break character determines direction** variance-reducing breaks (INDPRO 1984) produce inverse Lucas: LSR$<$1 for 7/9 models.
5. **LSR and absolute RMSE diverge** MSNN best absolute post-break RMSE despite high LSR; both metrics are needed.

Policy Implication

ML does *not* resolve the Lucas Critique. Flexible models (XGB) may degrade *more*. **HMM-family models** preferred where structural stability matters more than in-sample fit.

Headline (Paper Abstract)

"ML models degrade similarly to traditional models but still produce better forecasts under regime change" the MSNN's smooth extrapolation achieves best absolute post-break accuracy.

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